

Fusion of Machine Learning and Cryptography for Fast Data Encryption through Information Block Encoding

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1 Abstract

This project addresses the intrinsic trade-off between computational complexity and security in image encryption. By fusing machine learning with chaotic cryptography, the proposed system classifies plaintext data blocks based on information density to apply adaptive encryption levels. High-information blocks receive robust encryption, while moderate blocks utilize lightweight methods, significantly reducing overall encryption time without compromising data integrity[cite: 1].

2 Problem Statement

Protecting digital images often requires extensive mathematical operations to ensure robust security. However, this computational burden often makes real-time applications unfeasible, particularly in resource-constrained environments like IoT, cloud systems, and mobile applications[cite: 2, 9]. There is a critical need for an intelligent encryption framework that maintains high confidentiality and integrity while optimizing performance and reducing resource usage[cite: 6, 7].

3 Architecture

The proposed encryption technique follows a multi-stage adaptive framework:

- **Bit-Plane Decomposition:** Each pixel block is split into its eight bit-planes[cite: 10].
- **Rotation Phase:** Individual bit-planes are rotated at angles of 90° , 180° , and 270° to initiate the confusion process[cite: 10].
- **Chaotic Sequence Generation:** A chaotic map generates six random sequences using specific initial parameters $(L_{11}, L_{12}, L_{21}, L_{22}, s, K)$ [cite: 11].
- **Confusion Operation:** The generated sequences permute the bit-values across the rotated bit-planes, making unauthorized decryption challenging[cite: 12].
- **Diffusion Phase:** Modified bit-planes are recombined into a single pixel block. A 2D noisy image generated via chaotic maps is then XORed with the substituted image to produce the final ciphered result[cite: 13, 14].

4 Tools and Libraries

The implementation was developed using Python and the following key libraries[cite: 2]:

- **OpenCV (cv2):** For image processing and bitwise XOR operations[cite: 2].
- **NumPy:** For efficient array manipulation and mathematical operations on bit-planes[cite: 2].
- **Matplotlib & Seaborn:** For data visualization and displaying ciphered images[cite: 2].
- **PIL (Pillow):** For advanced image handling.
- **Pandas:** For managing datasets and performance metrics[cite: 2].

5 Machine Learning Models

The system utilizes supervised machine learning models to classify plaintext data blocks[cite: 4]:

- **Classification Task:** Models are trained to distinguish between high-density and moderate-density information blocks.
- **Adaptive Logic:** Based on the classifier's output, the framework dynamically selects the encryption strength, applying strong chaotic maps to sensitive areas and lightweight operations to less sensitive blocks[cite: 5].

6 Results and Performance

The system was evaluated using real-world datasets and compared against conventional full-encryption methods:

- **Efficiency:** A notable reduction in overall encryption time was achieved by avoiding over-encryption of moderate blocks[cite: 6].
- **Security:** Despite the optimization, the system maintains strong data confidentiality. The use of chaotic maps and bitwise XOR ensures high resistance to ciphertext-only and known-plaintext attacks[cite: 14].
- **Visual Output:** The final ciphered images demonstrate high entropy and successful noise diffusion, effectively masking the original image content[cite: 2].

7 Conclusion

This research demonstrates that integrating machine learning with cryptographic techniques can set a new benchmark for secure data communication, particularly in IoT contexts where speed and security are both essential[cite: 16, 17].